Comprehensive Diabetes Risk Analysis

From raw data and feature engineering to clinical insight

47,541
patients analysed

50
engineered features

10
questions answered

Five questions, in which they matter

01

Foundation

Can we trust the data?

27.9% of rows failed a check

02

Demographics

Who is at risk?

+11.8 points by age 60

03

Clinical

What is happening inside the body?

87% high risk above 140 mg/dL

04

Lifestyle

Which behaviours drive it?

A 12-point cliff, not a slope

05

Countermeasure

Can we change it?

12 earned back vs 9.7 inherited

The punchline: *genetics set the starting line - lifestyle moves it further than genetics did.*

PART 1 - FOUNDATION

Before analysis, Cleaning

13,257

rows quarantined - 27.9% of the file

Every headline is calculated on the 34,284 patients that survived the quality screen - not on the raw file.

Missing data

Height and weight recovered mathematically from BMI; median imputation for lipids and lifestyle metrics; 156 records left null.

Medical anomalies

3,797 blood-pressure readings were physically impossible (diastolic ≥ systolic). 3,692 BMI values sat outside human range.

Clinical categorisation

Seven age bands, plus standardised BMI, HbA1c, fasting-sugar and blood-pressure tiers for like-for-like comparison.

Custom risk indices

Unhealthy Habits (0-7) across diet, activity and sleep; Condition Count (0-5) across hypertension, heart disease and more.

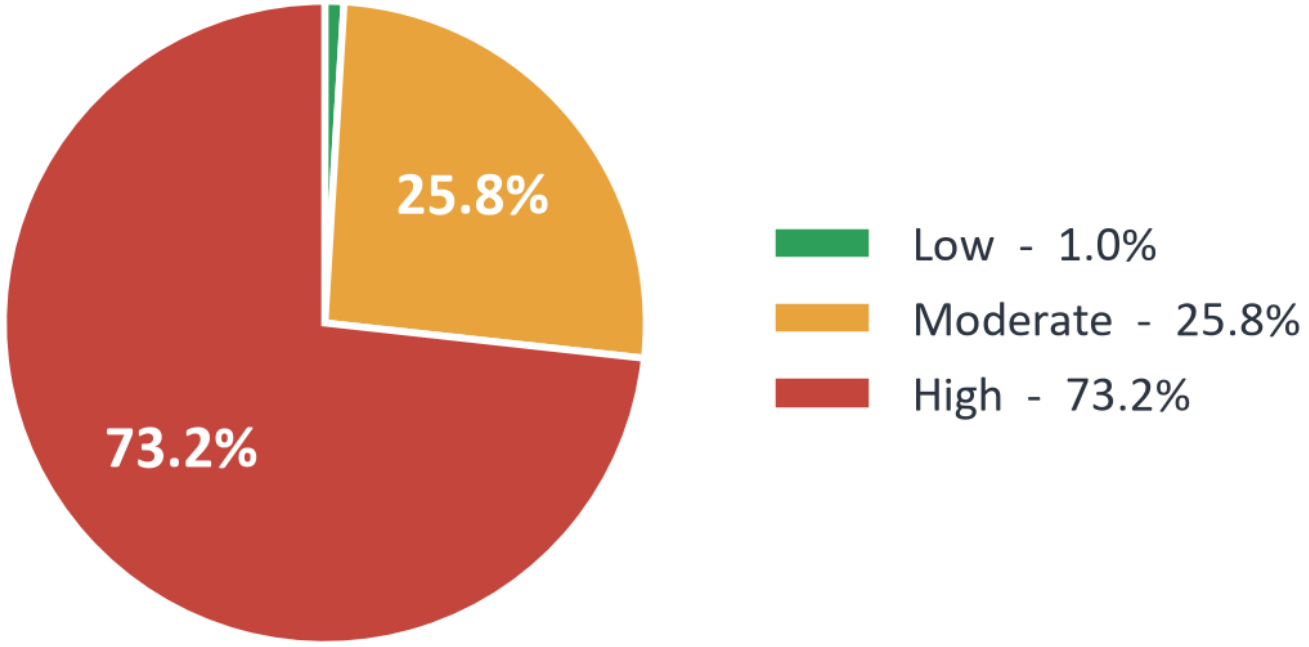
PART 1 - QUESTION 1 **What is the overall distribution of diabetes risk among patients?**

The baseline is already alarming

73.2%

of screened patients are already high risk

- 25,113 patients sit in the High band - roughly three in four.
- Low risk has all but vanished: 328 people, or 1.0%.

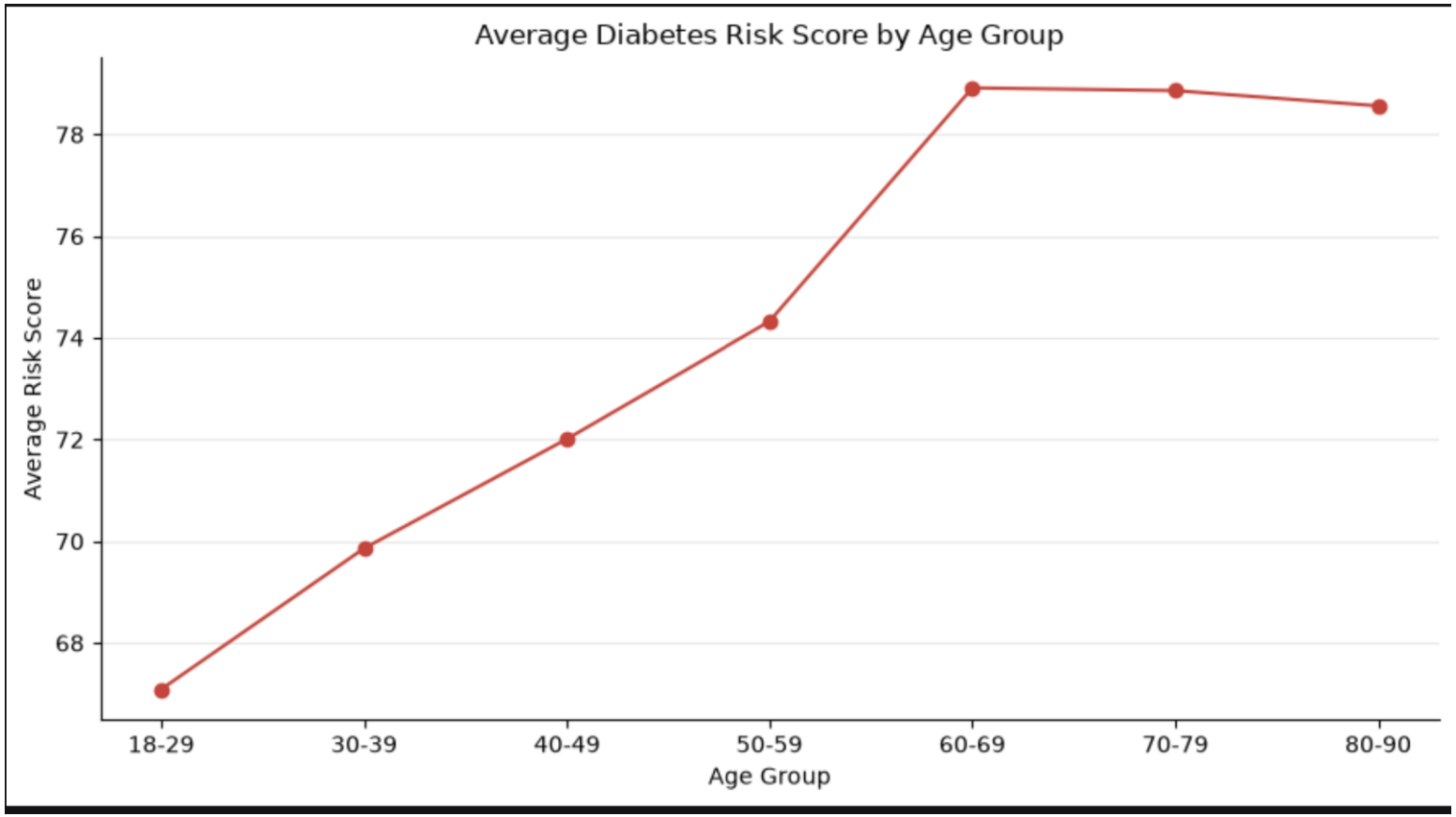


Risk climbs with age - then stops

+11

points of risk score from youngest to peak

- Average score rises steadily from 67 at 18-29 to 78 at 60-69.
- After 60 the curve flattens - the 70s and 80s add nothing.
- The accumulation happens in mid-life, not old age.

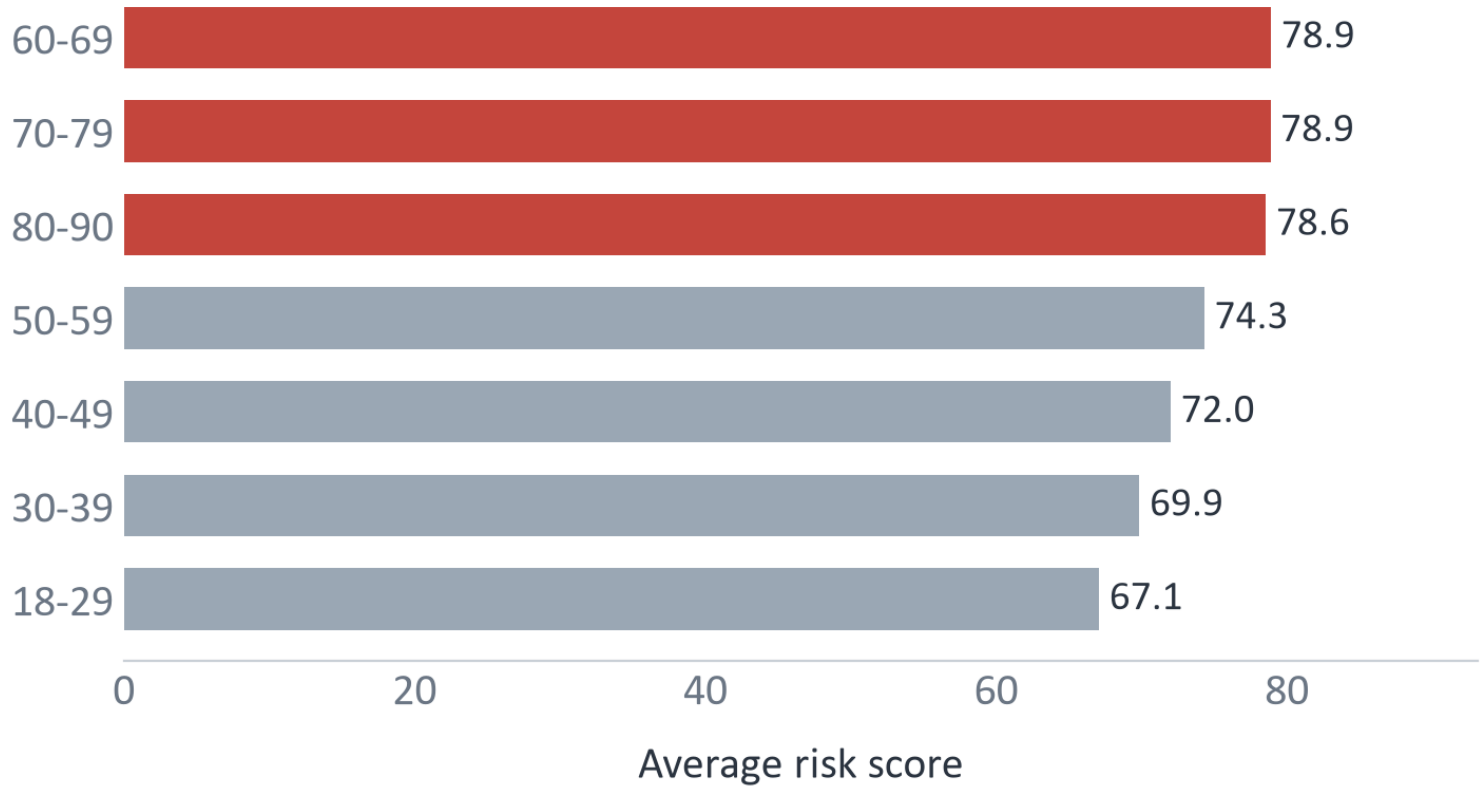


Three age bands share the top spot

78.9

average risk score across the 60+ bands

- 60-69, 70-79 and 80-90 are statistically indistinguishable.
- 18-29 sits lowest at 67.1 - but that is still above the Moderate line.
- Total spread across six decades of life is under 12 points.



HbA1c is the sharpest divider we have

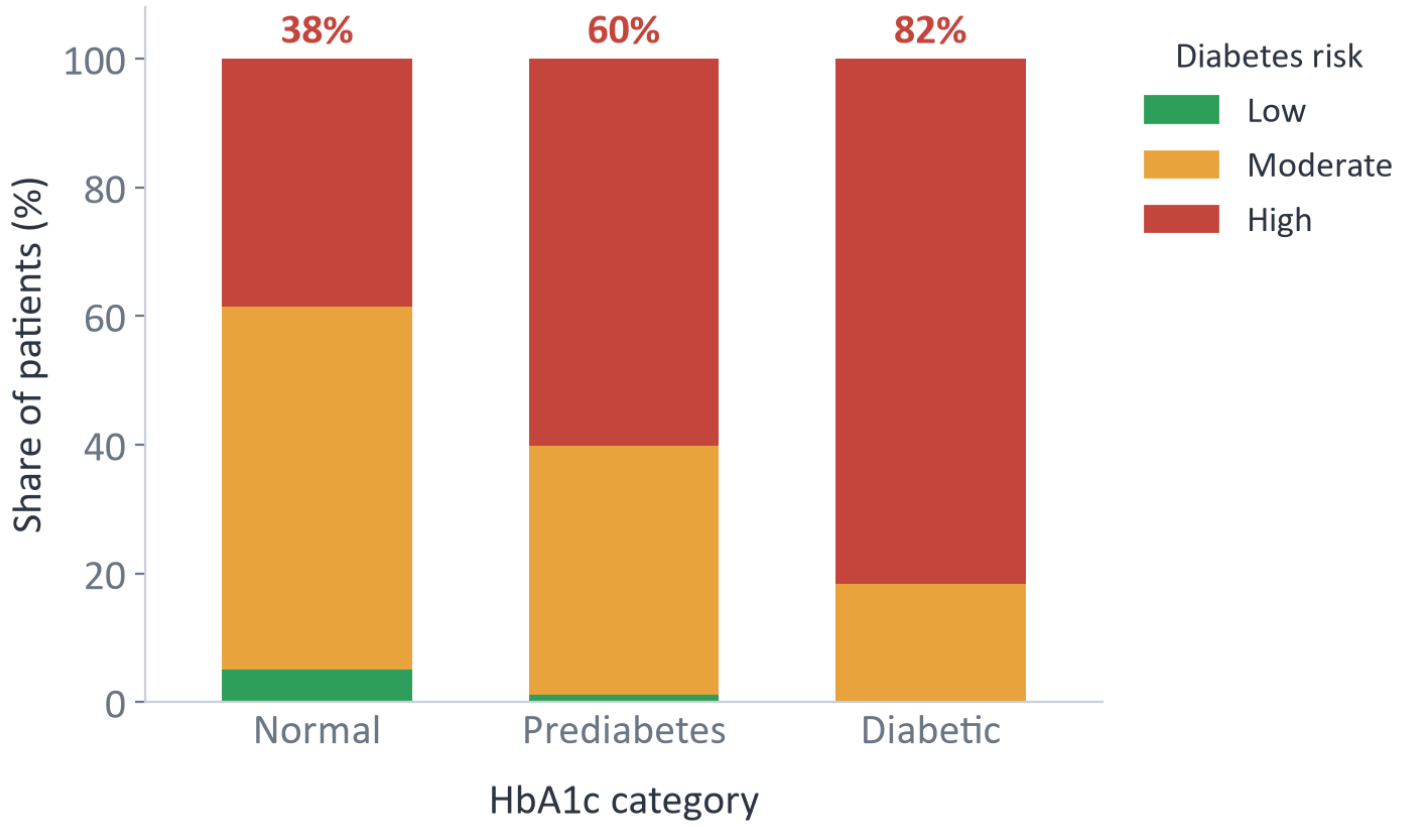
2.1x

more high-risk patients in the diabetic tier vs normal

- High-risk share: 38% Normal, 60% Prediabetes, 82% Diabetic.

- Each clinical tier adds roughly twenty points of risk.

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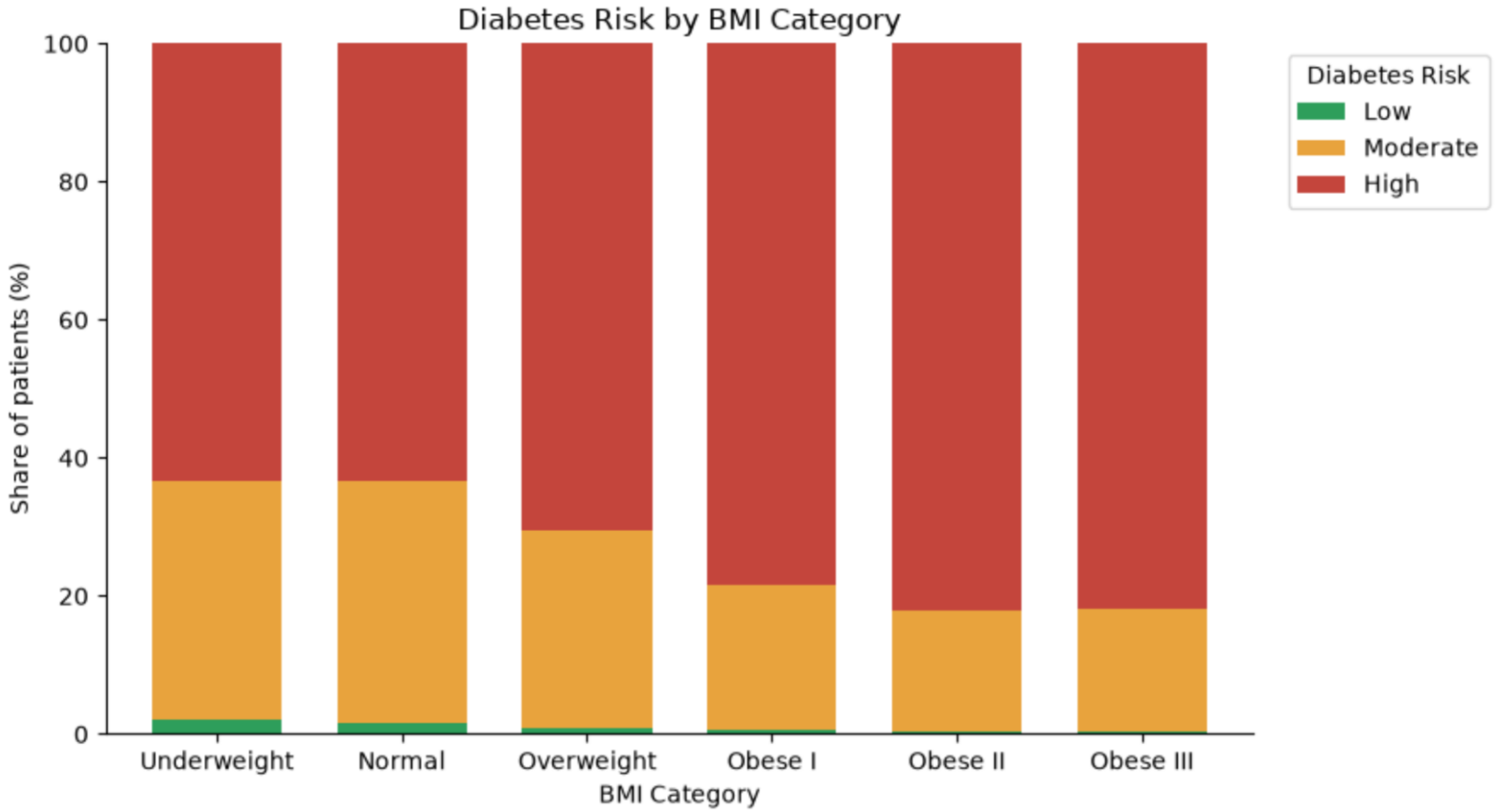


Weight moves the needle - up to a point

+18.8

point jump in high-risk share, Normal to Obese II

- High-risk share climbs from Normal to 80% at Obese II.
- Obese II and Obese III are level - the curve saturates.
- Underweight tracks Normal.

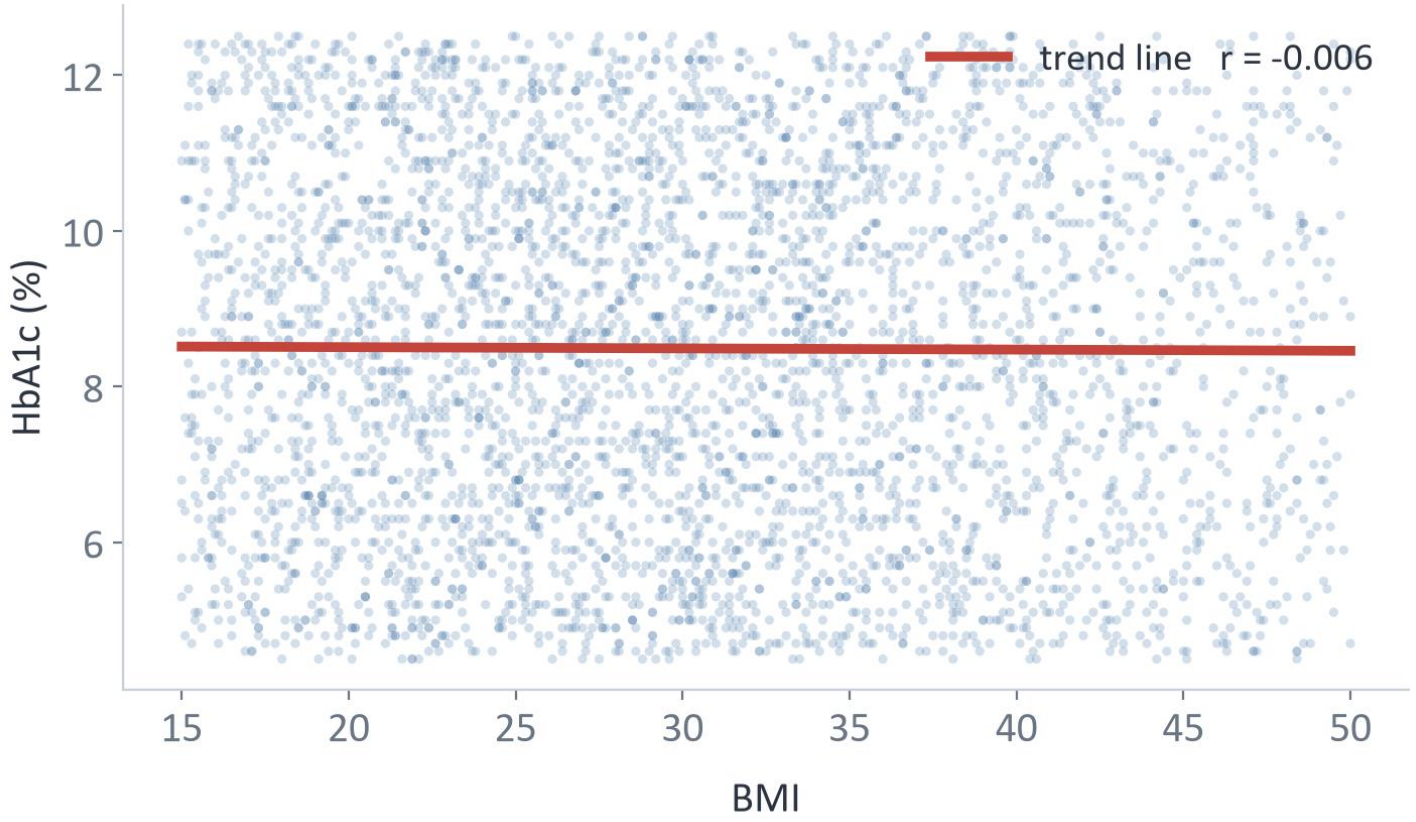


The assumption that did not survive

r = -0.006

correlation between BMI and HbA1c

- We expected a clear upward slope. The trend line is flat.
- Knowing a patient's BMI tells you nothing about their HbA1c here.
- Clinically these should correlate.

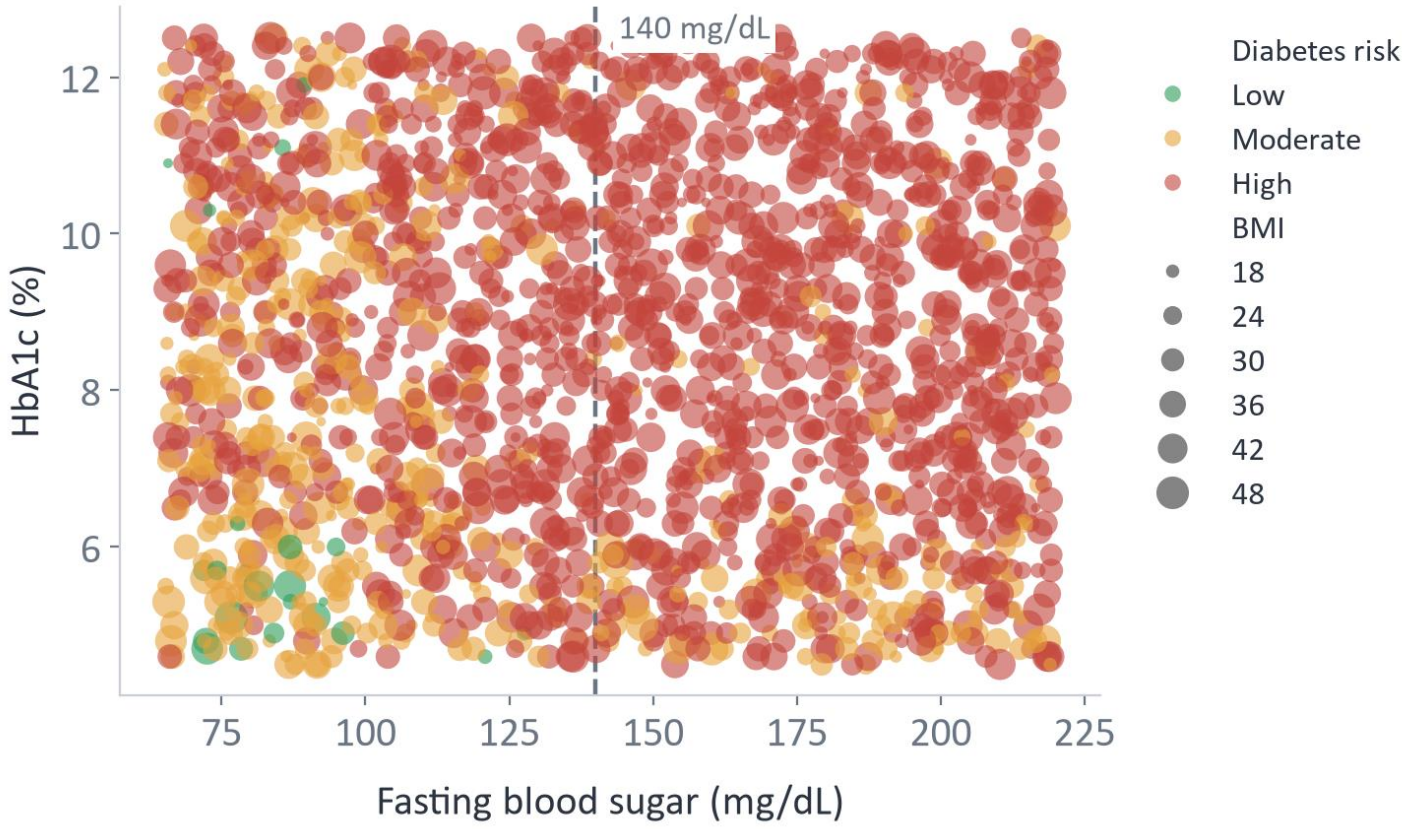


Mapping the danger zone

87.3%

high risk above 140 mg/dL fasting blood sugar (high risk count / total zone)

- Below 100 mg/dL that share falls .
- Ranked by strength: fasting blood sugar , HbA1c , BMI .

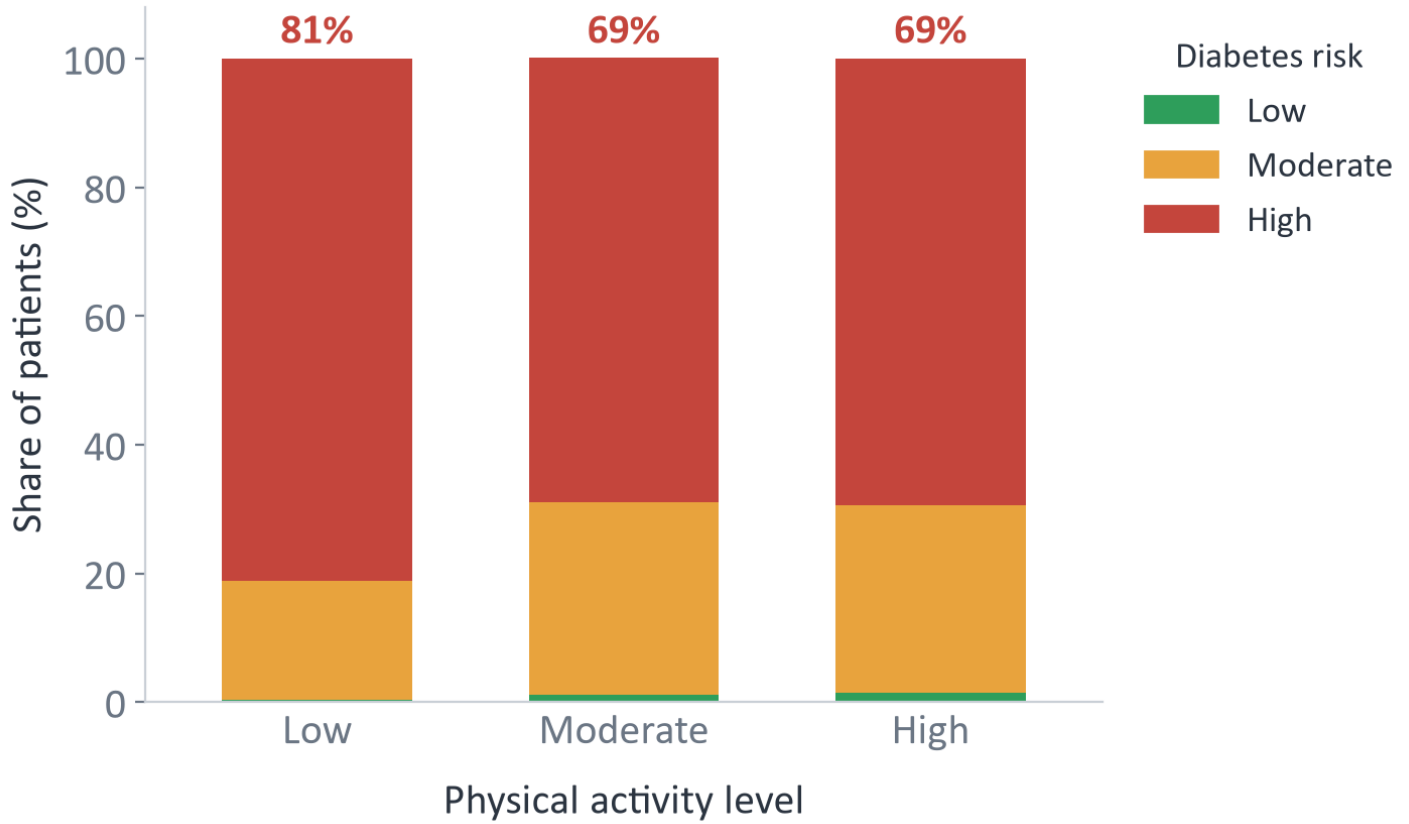


Daily Behaviors impacts

81.%

of low-activity patients are high risk

- Moderate activity: 69% high risk. High activity: 69%.
- Moving at all buys the full benefit .
- The 12-point penalty falls entirely on the inactive group.

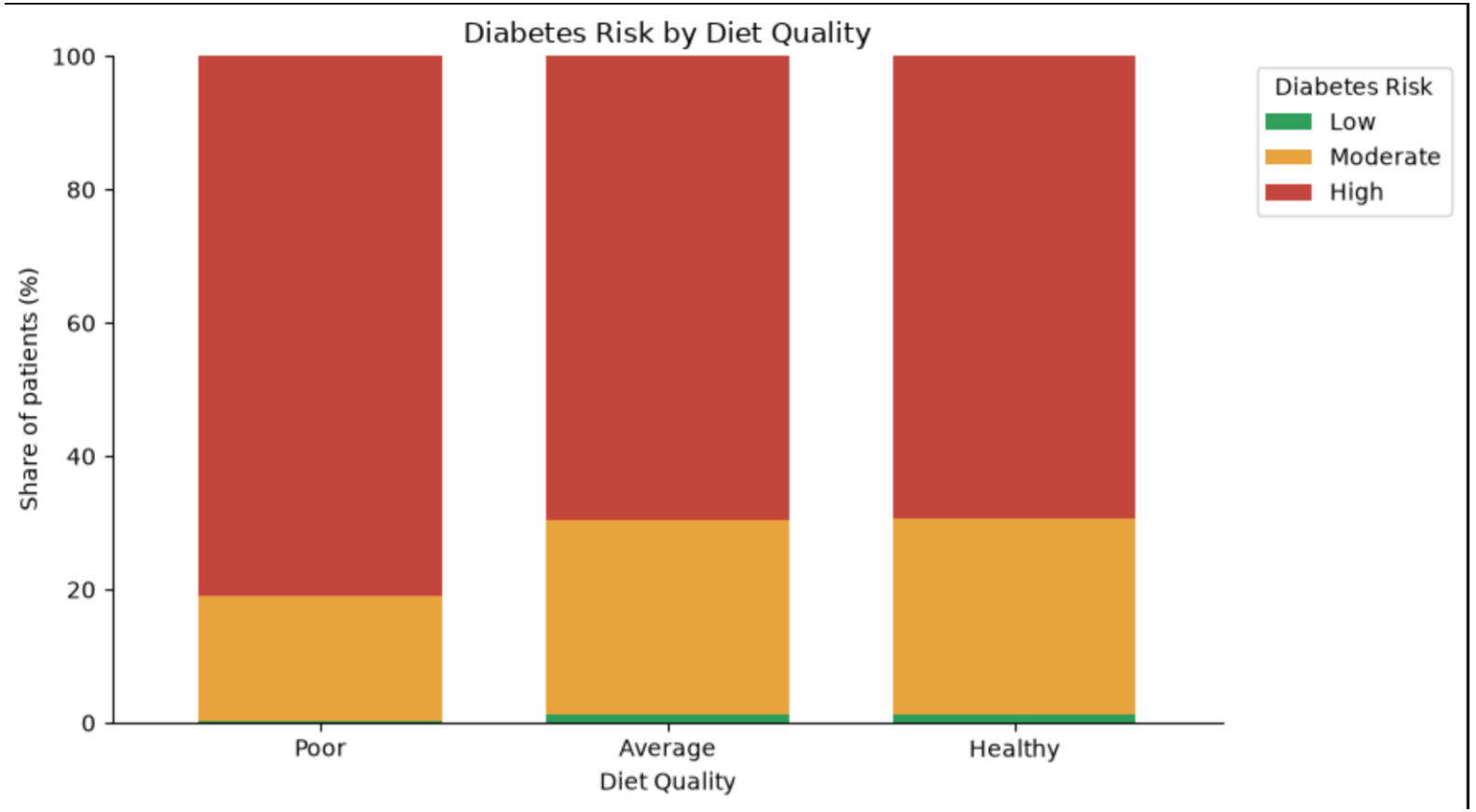


Diet tells the same story, twice

80%

of poor-diet patients are high risk

- Average diet: 70%. Healthy diet: 70%.
- Two independent behaviours, one identical shape.

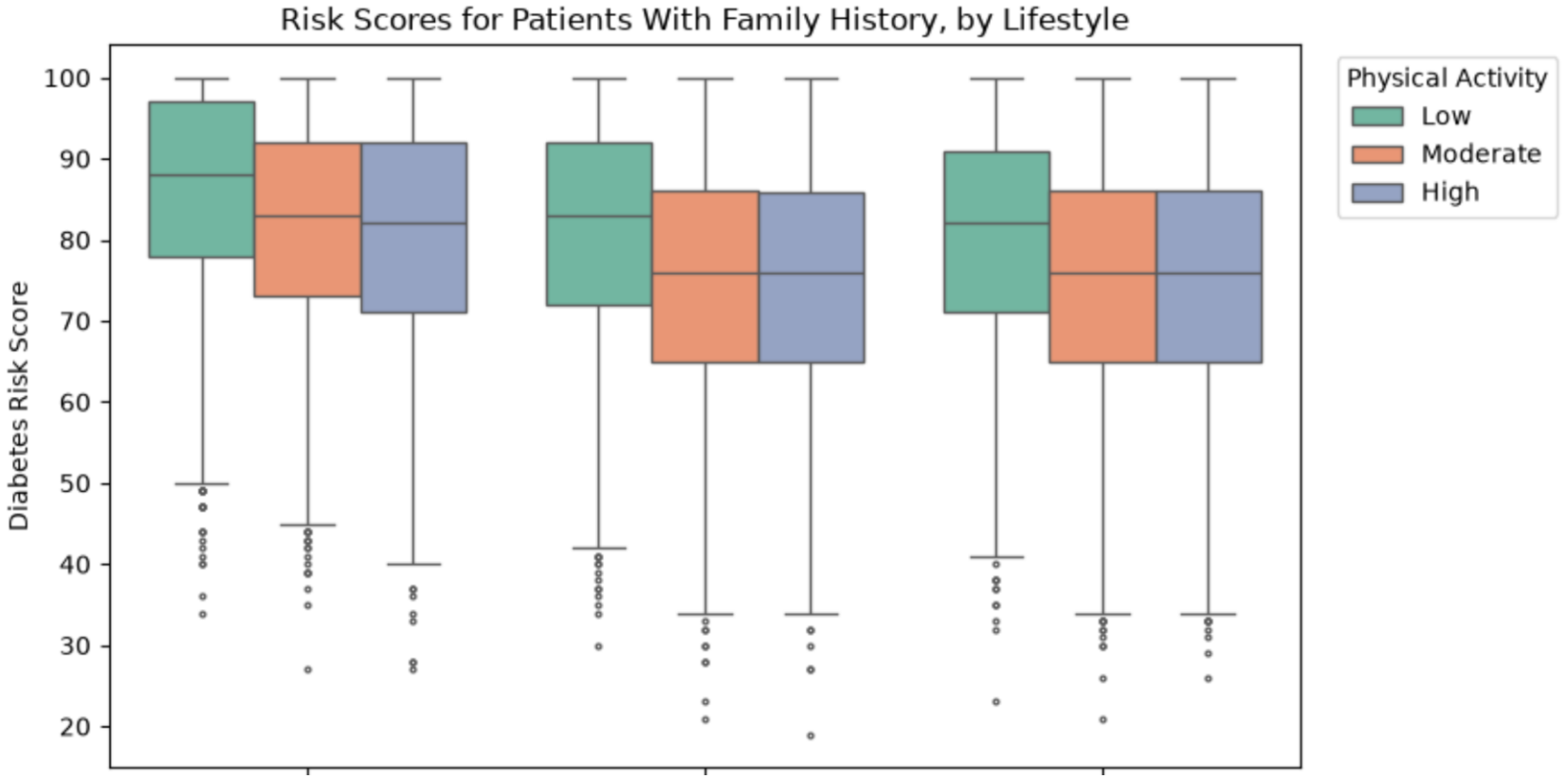


Genetics load the gun. Lifestyle pulls it back.

12 pts

median spread from worst to best lifestyle

- Family history costs 9.7 points on average: 78.9 vs 69.2.
- Within that group, lifestyle spans 88 down to 76 - a 12-point swing.
- Lifestyle moves risk further than the inherited penalty itself.



Three decisions this data supports

01

Screen on fasting glucose first

It separates risk groups roughly twice as strongly as BMI. If the budget covers one test per patient, this is the one that earns its cost.

02

Intervene in your 40-59

Risk accumulates through mid-life and plateaus at 60. Screening programmes aimed at the elderly arrive after the score is already set.

03

Exercise and regulated diet

Average and healthy behaviour score identically. The entire measurable benefit sits in moving people out of the poor category.

All three are cheaper than the programmes they replace.

CLOSING

Thank you - questions welcome, especially the awkward ones.

“Genetics load the gun. Lifestyle pulls it back.”

9.7 points inherited.

12 points earned back.